

travel booking

Software Requirements Specification

travel booking

Document Control

Version

Date

Author

Change Summary

1.0

2026-04-09

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Initial generated draft for structured review.

travel booking

Document ID: SRS-20260409-161644

Version: 1.0

Date: 2026-04-09

Author: usman asif

Req2Design · IEEE 830-1998 aligned · Generated output

INTRODUCTION

Purpose: To define the requirements for a travel booking system.

Scope: This document covers the development of a travel booking system which allows users to search, book, and manage their trips.

Definitions/Acronyms:

API: Application Programming Interface
E-ticket: Electronic ticket
FR: Functional Requirement
NFR: Non-functional Requirement
References: IEEE Std 830-1998

Overview: This system is designed to facilitate the planning and management of travel itineraries for individual travelers or groups.

OVERALL DESCRIPTION

Product Perspective: The product is a web-based travel booking system aimed at facilitating the entire journey from initial search to final departure.

Product Functions: Users can search for flights, hotels, car rentals, and other services; compare prices and conditions; make reservations; and manage their bookings.

User Characteristics: Primary users include frequent travelers, families, and business professionals who need assistance in organizing complex travel plans involving multiple destinations.

Constraints: The system must integrate with various external providers' APIs, support multiple payment methods, and comply with security standards.

Assumptions/Dependencies: Users have reliable internet connections and modern devices capable of running the application. Third-party service providers adhere to agreed-upon SLAs.

SPECIFIC REQUIREMENTS

External Interface Requirements

User Interface: The system provides intuitive and responsive interfaces for desktop and mobile devices. Hardware Interface: Compatible with standard personal computers and tablets. Software Interface: Supports integration with popular browsers and operating systems. Communication Interface: Capable of communicating securely over HTTPS protocols.

Functional Requirements

-

Input: Departure city, arrival city, dates, number of passengers, class preference.

-

Processing: Retrieve available flights from integrated APIs, calculate total cost based on selected criteria.

-

Output: Display list of suitable itineraries along with detailed pricing information.

-

Priority: High

FR-02 Compare Fares

-

Input: Selected itineraries from previous step.

-

Processing: Aggregate fare details from chosen routes, highlight differences in price, duration, layovers.

-

Output: Present comparison table showing key attributes side by side.

-

Priority: Medium

FR-03 Filter By Baggage And Refund Policies

-

Input: Selected itinerary.

-

Processing: Fetch relevant policy descriptions from provider databases.

-

Output: Show summary of baggage allowances and refund terms applicable to the booked tickets.

-

Priority: Low

FR-04 Reserve Seats

-

Input: Chosen seat numbers, passenger names.

-

Processing: Submit reservation request to airline database.

-

Output: Confirmation message indicating success or failure.

-

Priority: High

FR-05 Add Insurance

-

Input: Passenger details, optional coverage types.

-

Processing: Generate quote and purchase confirmation.

-

Output: Receipt detailing purchased insurance plan(s).

-

Priority: Medium

FR-06 Receive E-Ticket

-

Input: Reservation reference number.

-

Processing: Retrieve electronic ticket from carrier's server.

-

Output: Send email containing PDF version of e-ticket to customer.

-

Priority: High

FR-07 Handle Voucher Redemption

-

Input: Customer ID, redemption code.

-

Processing: Validate code against stored records, update account balance if valid.

-

Output: Notification confirming successful redemption or error message.

-

Priority: Medium

FR-08 Manage Fare Changes

-

Input: Booking reference, new date preferences.

-

Processing: Contact original supplier for updated rates, process change request.

-

Output: Updated itinerary reflecting revised schedule and costs.

-

Priority: High

FR-09 Offer Alternative Options During Checkout

-

Input: Current selection, expiration warning.

-

Processing: Provide suggestions for similar trips with remaining inventory.

-

Output: List of alternate offers with brief descriptions and estimated savings.

-

Priority: Medium

Non-functional Requirements

Usability: The system shall provide easy-to-use interfaces that enable users to perform tasks efficiently without requiring extensive training.

Reliability: The system shall ensure high availability and minimal downtime through redundant systems and regular maintenance schedules.

Performance: The system shall respond within 5 seconds for critical operations.

Portability: The system shall be deployable across different platforms including web and mobile applications.

SYSTEM FEATURES

Feature 1: Flight Search & Comparison

Allows users to find and compare flight options between multiple cities. Includes real-time updates and dynamic pricing adjustments.

Feature 2: Hotel Booking Integration

Enables users to select accommodations near airports or preferred locations. Provides ratings and reviews from trusted sources.

Feature 3: Car Rental Services

Facilitates rental agreements directly through partner agencies. Offers flexible pick-up/drop-off times and locations.

Feature 4: Travel Insurance

Bundles comprehensive protection packages tailored to specific needs. Automatically integrates with existing reservations for seamless coverage.

Feature 5: Trip Management Tools

Tracks ongoing journeys via GPS tracking technology. Sends automated alerts about delays or cancellations.

Feature 6: Secure Payment Gateway Support

Processes payments safely through certified financial institutions. Ensures compliance with PCI DSS regulations.

Feature 7: Customer Profile Storage

Maintains personalized information for quicker future bookings. Protects sensitive data with encryption techniques.

Feature 8: Reminders & Notifications

Delivers timely messages regarding upcoming departures or important deadlines. Customizable settings allow users to tailor notification frequency and type.

Feature 9: Simple Change/Cancellation Policy

Simplifies the process of modifying or canceling existing bookings. Applies standardized rules consistently across all transaction types.

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